

Introductions

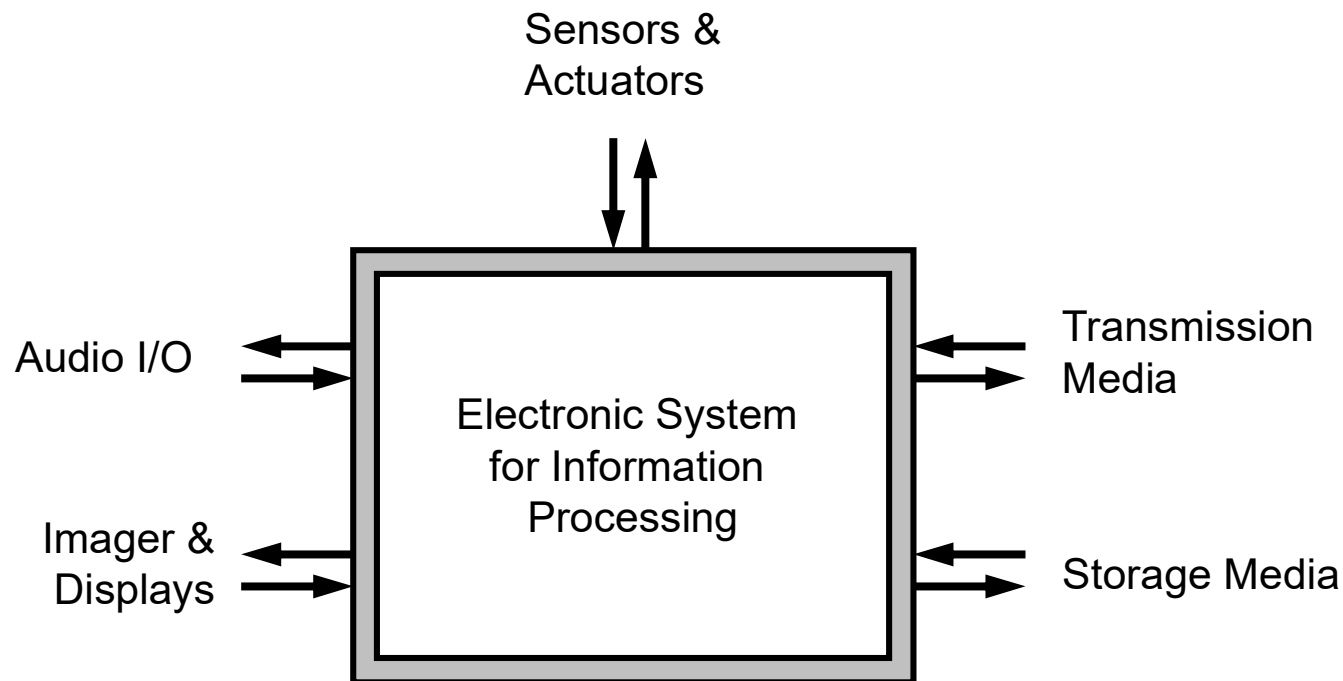
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Teaching Staffs

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Motivation (1)

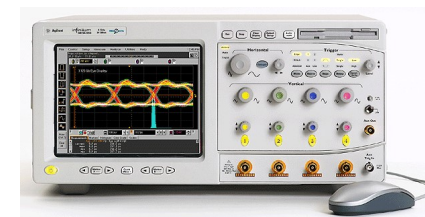
- Information is increasingly being processed in the digital domain; yet, the real world is analog by nature
 - Whenever an electronic device wants to talk to the real world, it needs a data converter.



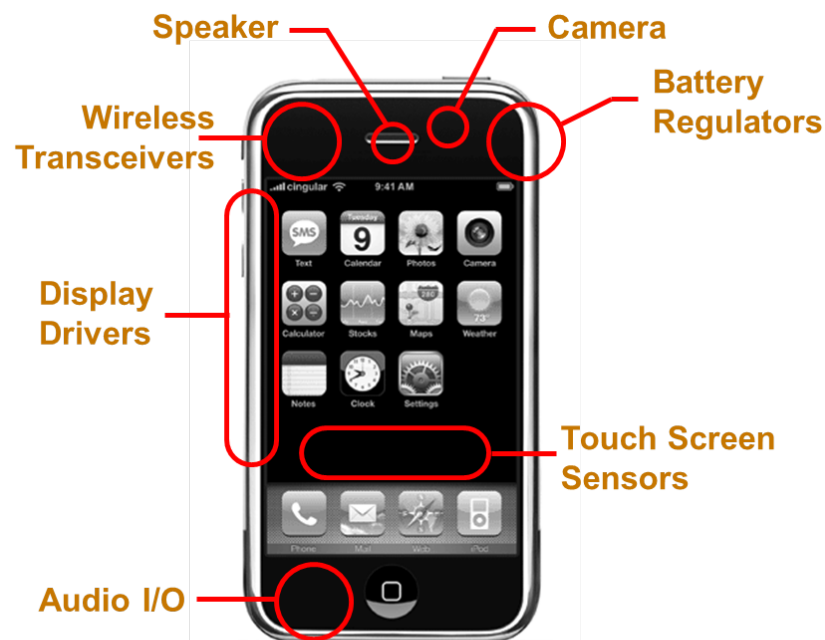
Motivation (2)

- Benefits of digital signal processing
 - Amenable to automated design and test
 - Enhanced functionality and flexibility
 - Direct benefit from the scaling of VLSI technology
 - Reduced sensitivity to "analog" noise
 - "Arbitrary" precision
- Issues
 - Data converters are difficult to design
 - Especially due to ever-increasing performance requirements
 - Data converters often present a performance bottleneck
 - Speed, resolution or power dissipation of the A/D or D/A converter can limit overall system performance

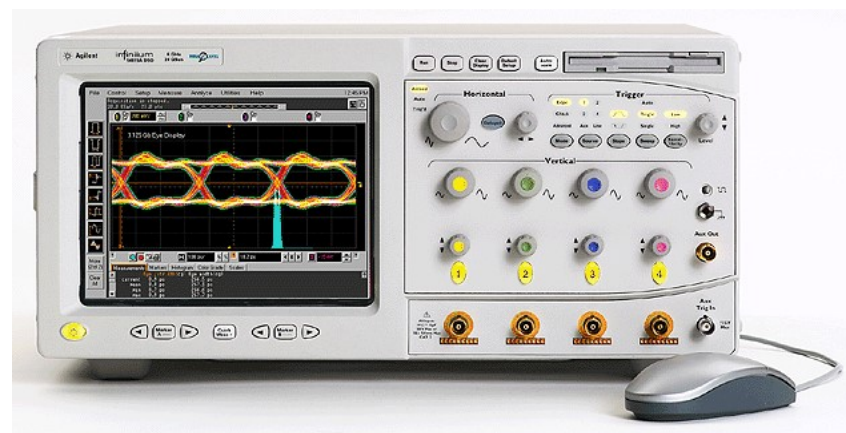
Data Converter Applications



Examples

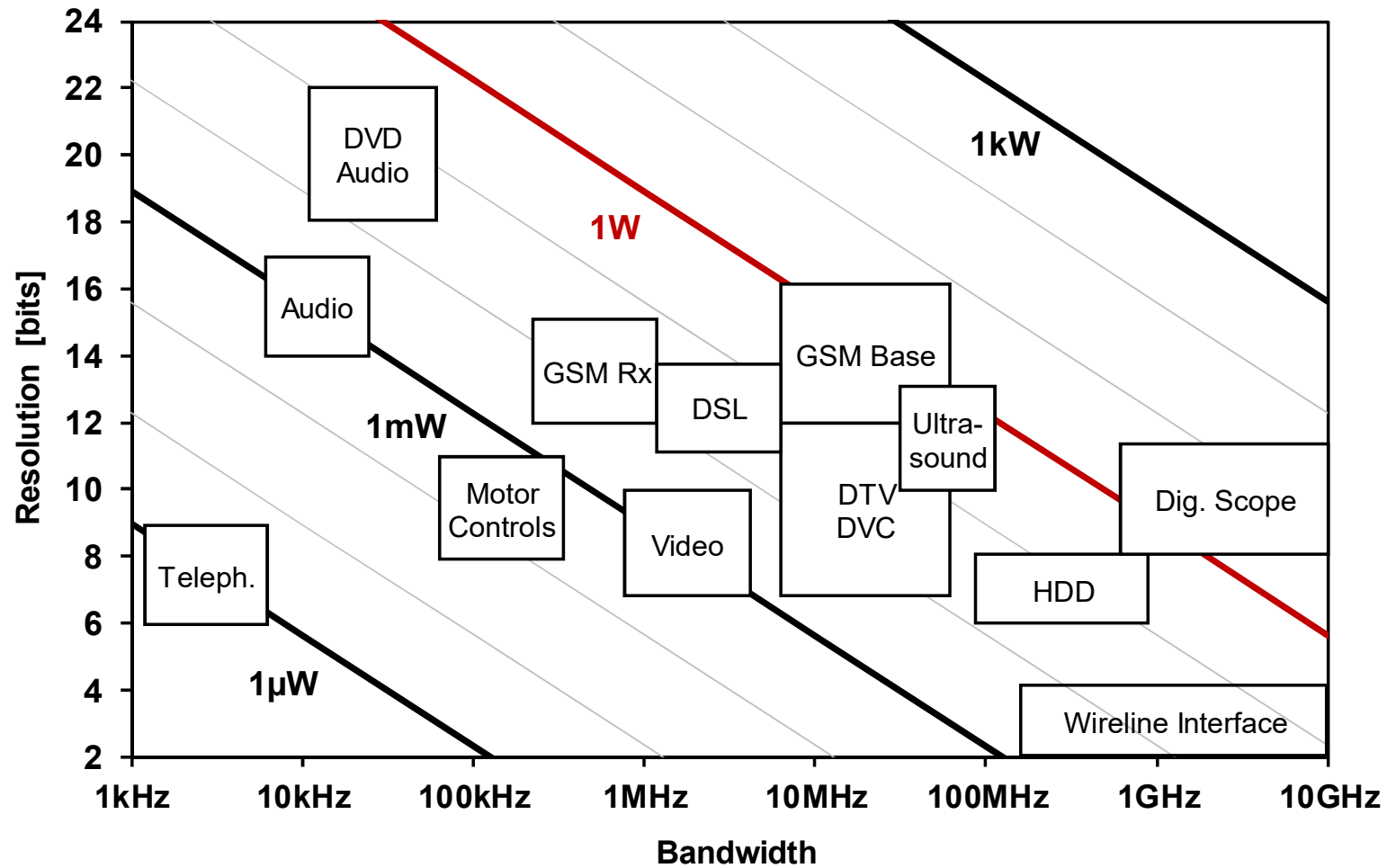


More than 20 data converters are used in each iPhone system

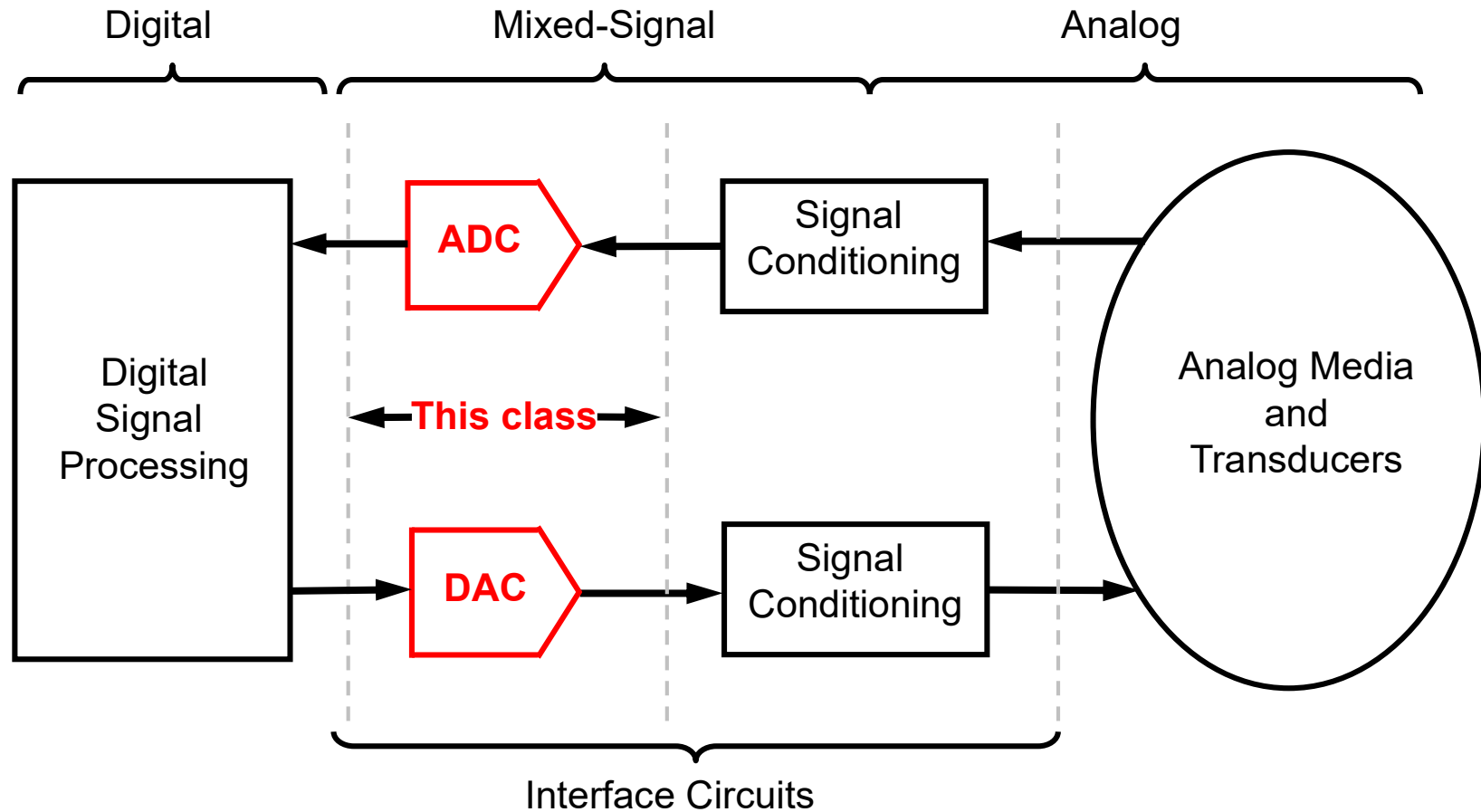


- A typical cell phone contains more than 20 data converter
- Digital oscilloscopes are essentially high-performance ADCs

A/D Converter Application Space



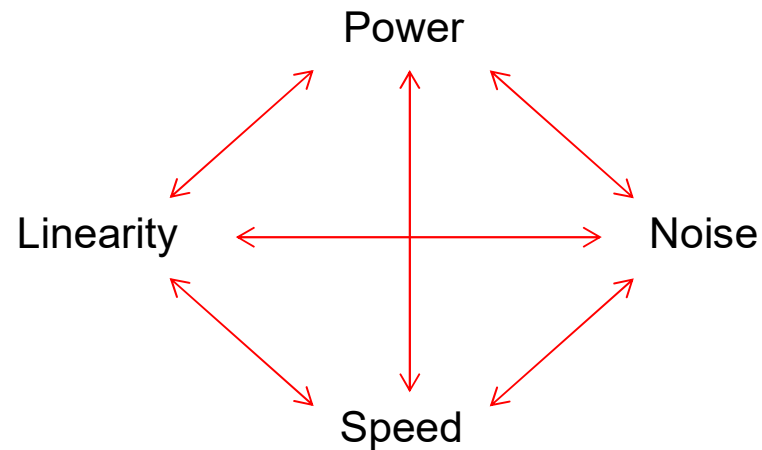
Course Contents



How I look at data converters



Crown in analog mixed-signal IC design
Difficult but fun!



Course Prerequisites

- Undergraduate electronic circuit courses:
 - Device physics and models
 - Transistor level analog circuits
 - Frequency response, feedback, and noise
- Graduate circuit course:
 - Analog IC design
- Prior exposure to SPICE (Cadence) and Matlab

Course Topics

- Ideal sampling, reconstruction, & quantization
- Sampling circuits
- Voltage comparators
- Nyquist-rate ADCs and DACs
- Oversampled ADCs and DACs
- Data converter performance trends and limits
- Data converter testing

Let us go over the syllabus
